

-- Cox Regression - Sarcopenia Onset -----

i Definition : ewgsop2
i Covariates : fixed
i Dairy : continuous
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
Fitting .imp = 1 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 2 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 3 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 52 events
Fitting .imp = 6 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 7 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 8 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 9 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 10 / 20 ...
Events: 58 sarcopenia cases out of 1121 participants

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 12 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 13 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 14 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 15 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 16 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

Fitting .imp = 18 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 19 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 20 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events

-- Pooled adjusted model results (Rubin's rules) --

A tibble: 8 x 5 term HR CI_low CI_high p.value <chr> <dbl> <dbl> <dbl> <dbl>
1 dairy_exposure 1.000 0.998 1.00 0.840 2 BMI 0.692 0.617 0.776 0.000000128 3
education_levelMedium (ISCED 3-4) 1.01 0.445 2.30 0.978 4 education_levelHigh
(ISCED 5-8) 1.02 0.325 3.19 0.975 5 smoking_statusFormer 0.899 0.474 1.70 0.737
6 smoking_statusCurrent 0.788 0.296 2.10 0.626 7 mvpa_min_day_f1 0.999 0.996
1.00 0.495 8 diabetes_statusDiabetes 1.48 0.326 6.76 0.602

-- 1. Proportional Hazards - cox.zph() | pass: global p > 0.05

```
chisq df p dairy_exposure 0.78243 1 0.376 BMI 0.00295 1 0.957 education_level
5.20875 2 0.074 smoking_status 4.90722 2 0.086 mvpa_min_day_f1 0.35718 1 0.550
diabetes_status 1.71405 1 0.190 GLOBAL 11.60616 8 0.170
??? PH assumption not violated (global p = 0.1697)
Schoenfeld plots saved to
03_outputs/Cox/mice_ewgsop2_fixed_continuous_27072026_1254
```

-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear

```
'geom_smooth()' using formula = 'y ~ x'
'geom_smooth()' using formula = 'y ~ x'
'geom_smooth()' using formula = 'y ~ x'
Linearity plots saved to
03_outputs/Cox/mice_ewgsop2_fixed_continuous_27072026_1254
```

-- 3. Outliers - deviance & martingale residuals | pass: |dev| <= 3

i 4 observation(s) with |deviance residual| > 3.

-- 4. Influential observations - std. DFBetas | pass: no extreme spikes

i 27 obs exceed |DFBeta_std| > 0.2.

-- 5. Overall Fit - Cox-Snell residuals | pass: points on 45° diagonal

```
Cox-Snell plot saved to
03_outputs/Cox/mice_ewgsop2_fixed_continuous_27072026_1254
```

-- 6. Multicollinearity - VIF | pass: VIF < 5 (< 10 relaxed)

```
GVIF Df GVIF^(1/(2*Df)) dairy_exposure 1.068 1 1.033 BMI 1.144 1 1.070
education_level 1.261 2 1.060 smoking_status 1.152 2 1.036 mvpa_min_day_f1
1.052 1 1.026 diabetes_status 1.077 1 1.038
```

-- Incidence rates by dairy quartile [mice]

quartile	n_pts	n_events	person_years	rate_per_1000py	ci_low	ci_high	1	Q1 (low)							
281	10.00	19904.70	0.50	0.24	0.92	2	Q2	280	9.70	19833.92	0.49	0.23	0.91	3	Q3
280	13.00	19898.40	0.65	0.35	1.12	4	Q4 (high)	280	17.45	19946.00	0.87	0.51	1.39		

Incidence table saved to
03_outputs/Cox/mice_ewgsop2_fixed_continuous_27072026_1254
Model results saved to
03_outputs/Cox/mice_ewgsop2_fixed_continuous_27072026_1254/mice_model_results.csv